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MILITARY SPECIFICATION

COILS, TUBE DEFLECTION; AND COILS, TUBE FOCUSING

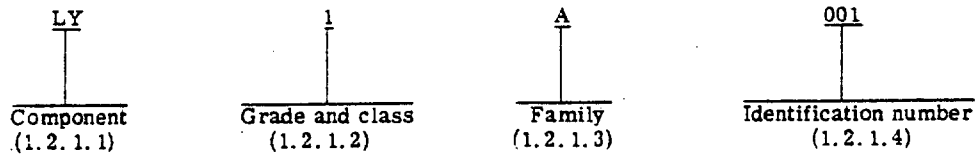
This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE.

1.1 Scope. This specification covers the general requirements for deflection coils and focusing coils used in conjunction with cathode-ray tubes for magnetic deflection and magnetic focusing. Procurement of deflection coils and focusing coils of a specific design will require additional data in the form of complementary Service documents, giving detailed electrical and mechanical requirements and applicable additions and exceptions to the general requirements and tests specified herein (see 3.1).

1.2 Classification.

1.2.1 Type designation. The type designation shall be in the following form, and as specified (see 3.1 and 6.1):



1.2.1.1 Component. Deflection coils and focusing coils are identified by the two-letter symbol "LY".

1.2.1.2 Grade and class. The grade and class are identified by a single digit in accordance with table I.

Table I. Grade and class.

Symbol	Grade	Class
1	1	A
2	2	A
3	3	A
4	1	B
5	2	B
6	3	B

1.2.1.2.1 Grade. The grades, including their environmental characteristics, are as follows:

Grade 1 - Resistant to immersion and moisture. (See 4.7.13 and 4.7.14.)
Grade 2 - For use in sealed assemblies.
Grade 3 - Resistant to moisture only. (See 4.7.14.)

1.2.1.2.2 Class. The classes, denoting the maximum operating temperatures (temperature rise (see 4.7.5) plus maximum ambient temperature), are as follows:

Class A - 85°C.
Class B - 125°C.

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1.2.1.3 Family. The family is identified by a single letter, indicating its function, as follows:

A - Coil, rotating deflection.
B - Coil, nonrotating deflection.
C - Coil, focusing.

1.2.1.4 Identification number. The identification number indicates a specific deflection coil or focusing coil. The number, assigned serially when a coordinated detail specification is approved, comprises three digits. If the number is less than 10, it is preceded by two zeros; if it is more than 9 and less than 100, it is preceded by one zero. The number will be assigned by the Armed Services Electro-Standards Agency (see 6.3).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-585 - Boxes; Wood, Wirebound.
PPP-B-591 - Boxes, Fiberboard, Wood-Cleated.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Fiberboard.
PPP-B-665 - Boxes; Paperboard, Metal Stayed (Including Stay Material).
PPP-B-676 - Boxes, Set-Up, Paperboard.
PPP-T-76 - Tape, Pressure-Sensitive Adhesive Paper, Water Resistant (for Carton Sealing).
PPP-T-97 - Tape; Pressure-Sensitive Adhesive, Filament Reinforced.

MILITARY
MIL-P-116 - Preservation, Methods of.
MIL-L-10547 - Liners, Case, Waterproof.

STANDARDS

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U.S. Military Property.
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

NATIONAL BUREAU OF STANDARDS
Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D.C.)

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, One Park Avenue, at 33rd Street, New York 16, N.Y.)

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3. REQUIREMENTS

3.1 Detail requirements for individual types of deflection and focusing coils. - Detail requirements or exceptions applicable to individual types of coils shall be as specified in the applicable complementary Service document (eg, detail specification, MS military standard, figure, drawing, specification sheet, or other document). In the event of any conflict between requirements of this specification and the complementary Service documents, the latter shall govern (see 6.1).

3.2 Preproduction. - Coils furnished under this specification shall be product which has been tested, and has passed the preproduction inspection specified in 4.5 (see 6.2).

3.2.1 Information to be furnished with preproduction sample. - The applicable information outlined in 6.1 shall be furnished with the preproduction sample, together with any other pertinent information as required by the Government.

3.3 Material. - The material for each part shall be as specified herein; however, when a definite material is not specified, a material shall be used which will enable the coils to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.3.1 Substitution of material. - If the supplier desires to substitute another material for a specified material or fabricated part, he shall submit a statement to the Government describing the proposed substitution, together with evidence to substantiate his claims that such substitute is suitable. At the discretion of the Government, test samples may be required to prove the suitability of the proposed substitute. Before such substitutions are made, approval for each substitution shall be obtained in writing from the Government.

3.3.2 Flammable materials. - So far as practicable, materials used in the construction of deflection coils and focusing coils shall be nonflammable and nonexplosive.

3.3.3 Corrosive materials. - Corrosive materials used in any of the manufacturing processes shall be removed or neutralized so that no corrosion will result from such use. So far as practicable, materials used in the construction of coils shall be noncorrosive.

3.3.4 Screws, nuts, and washers. - All external screws, nuts, and washers shall be of corrosion-resistant material or shall be protected against corrosion.

3.3.5 Magnet wire. - Magnet wire shall be as specified (see 3.1 and 6.1.2).

3.3.6 Varnish impregnant. - Varnish impregnant shall be as specified (see 3.1 and 6.1.2).

3.4 Design and construction. - Coils shall be of the design, construction, and physical dimensions specified (see 3.1 and 6.1.2).

3.4.1 Weight. - The weight of the coil shall be as specified (see 3.1 and 6.1.2).

3.4.2 Mounting. - Coils shall be mounted concentrically and secured rigidly so as to prevent any change in the relative position of the elements of the coils. The type of mounting shall be as specified (see 3.1 and 6.1.2).

3.4.3 Threaded parts. - All threaded parts shall be in accordance with Handbook H28. Where practical, all threads shall be in conformity with the coarse thread series.

3.4.4 External can shields. - External can shields (when specified (see 3.1 and 6.1.2)) shall be so constructed and connected that they can be effectively mounted mechanically and grounded electrically by their normal mounting means.

3.4.5 Terminals. -

3.4.5.1 Solder terminals. - Solder terminals may be of any shape and shall be capable of being readily soldered.

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3.4.5.2 Winding ends. - The ends and end turns of layer-wound coils and the finish turns of universal-wound coils shall be secured in such a manner as to anchor them securely in place. The length of leads from windings shall be as specified (see 3.1 and 6.1.2), and when necessary shall be supported to prevent vibration.

3.5 Electrical characteristics. - When coils are tested as specified in 4.7.2 to 4.7.2.8, inclusive, the applicable electrical characteristics, with values and tolerances, shall be as specified (see 3.1 and 6.1.2).

3.6 Dielectric withstanding voltage. - When coils are tested as specified in 4.7.3, there shall be no arcing, flashover, breakdown of insulation, or other damage.

3.7 Insulation resistance (when applicable (see 3.1 and 6.1.2)). - When measured as specified in 4.7.4, the insulation resistance shall be 1,000 megohms, minimum, for grade 1 coils; and as specified (see 3.1 and 6.1.2), for grades 2 and 3 coils. Insulation resistance is normally applicable, except for single-winding coils without external can shields for mounting means.

3.8 Temperature rise (when applicable (see 3.1 and 6.1.2)). - When coils are tested as specified in 4.7.5, the allowable temperature rise of any winding above the maximum operating temperature specified (see 3.8.1) shall not exceed the value specified (see 3.1 and 6.1.2); there shall be no evidence of physical damage.

3.8.1 Temperature range. - The maximum operating-temperature range shall be -65° C to +85° C, inclusive, for class A coils, and -65° C to +125° C, inclusive, for class B coils.

3.8.2 Storage at reduced temperatures. - Coils shall meet the requirements of this specification after storage for 48 hours at a temperature of -62° ± 2° C.

3.9 Terminal strength. - When coils are tested as specified in 4.7.6, there shall be no loosening or rupturing of the terminals or other damage.

3.10 Effects of soldering. - When coils are tested as specified in 4.7.7, there shall be no softening of the insulation or loosening of the windings or terminals.

3.11 High temperature. - When coils are tested as specified in 4.7.8, the dielectric withstanding voltage, and insulation resistance (when applicable (see 3.1 and 6.1.2)) shall be as specified in 3.6 and 3.7, respectively; there shall be no softening of the insulation or loosening of the windings or terminals.

3.12 Polarity (focusing coils only). - When focusing coils are tested as specified in 4.7.9, the polarity shall be as specified (see 3.1 and 6.1.2).

3.13 Flux alinement (deflection coils only). - When deflection coils are tested as specified in 4.7.10, the flux alinement shall be as specified (see 3.1 and 6.1.2).

3.14 Vibration. - When coils are tested as specified in 4.7.11, the dielectric withstanding voltage, insulation resistance (when applicable (see 3.1 and 6.1.2)), and the electrical characteristics shall be as specified in 3.6, 3.7, and 3.5, respectively; there shall be no evidence of physical damage.

3.15 Thermal shock. - When coils are tested as specified in 4.7.12, the dielectric withstanding voltage, and insulation resistance (when applicable (see 3.1 and 6.1.2)) shall be as specified in 3.6 and 3.7, respectively; there shall be no excessive peeling, flaking, chipping, cracking, or other impairment of the protective coating, nor evidence of leakage or physical damage, or corrosion affecting the mechanical or electrical operation.

3.16 Immersion (grade 1 only). - When coils are tested as specified in 4.7.13, the dielectric withstanding voltage, and insulation resistance (when applicable (see 3.1 and 6.1.2)) shall be as specified in 3.6 and 3.7, respectively; there shall be no excessive peeling, flaking, chipping, cracking, or other impairment of the protective coating, nor evidence of leakage or physical damage, or corrosion affecting the mechanical or electrical operation.

3.17 Moisture resistance (grades 1 and 3 only). - When coils are tested as specified in 4.7.14, the dielectric withstanding voltage, insulation resistance (when applicable (see 3.1 and 6.1.2)), and the electrical characteristics shall be as specified in 3.6, 3.7, and 3.5, respectively; there shall be no excessive peeling, flaking, chipping, cracking, crazing, or other impairment of the protective coating, nor evidence of physical damage, or corrosion affecting the mechanical or electrical operation.

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3.18 Marking. - Coils shall be marked in accordance with Standard MIL-STD-130, with the type designation and the manufacturer's name, trademark, or code symbol, part number (when applicable), and any other information specified (see 3.1 and 6.1.2).

3.19 Workmanship. - Coils shall be processed in such a manner as to be uniform in quality and shall be free from defects that will affect life, serviceability, or appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. -

4.1.1 Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1.1 Test equipment and inspection facilities. - Test equipment and inspection facilities shall be of sufficient accuracy, quality, and quantity to permit performance of the required inspection. The supplier shall establish calibration of inspection equipment to the satisfaction of the Government.

4.2 Classification of inspection. - The examination and testing of coils shall be classified as follows:

(a) Component-materials inspection (see 4.3).
(b) Preproduction inspection (see 4.5).
(c) Acceptance inspection (see 4.6).
    1. Inspection of product for delivery (see 4.6.1).
    2. Inspection for preparation for delivery (see 4.6.2).

4.3 Component-materials inspection. - Component-materials inspection shall consist of verification that the component materials listed in table II, used in fabricating the coils, are in accordance with the applicable referenced specifications or requirements prior to such fabrication.

Table II. Component-materials inspection.

Component material	Requirement paragraph	Applicable specification
Magnet wire	3.3.5	As specified. (See 3.1 and 6.1.2.)
Varnish impregnant . . .	3.3.6	As specified. (See 3.1 and 6.1.2.)

4.4 Inspection conditions. - Unless otherwise specified herein, all inspection shall be made at an ambient temperature of 25° $^{+10°}_{-5°}$ C, and ambient relative humidity and pressure.

4.4.1 Test voltage. - The test voltage shall be sufficiently free of distortion and noise to such a degree as not to produce any false or spurious indications.

4.4.2 Test frequency. - When a test frequency is specified with no tolerance, the frequency used shall be within 1 percent of the specified value.

4.5 Preproduction inspection. - Preproduction inspection will be performed by the supplier, after award of contract and prior to production, at a location designated by the Government (see 6.1.). Unless otherwise specified (see 6.1), the sample and test routine shall be as specified in 4.5.1, 4.5.2, and table III. Preproduction approval is valid only on the contract under which it is granted, unless extended by the Government to another contract. (see 3.2 and 3.2.1).

4.5.1 Sample. - Four sample units of each type shall be submitted for preproduction inspection, and shall be representative of the materials, processes, and elements (eg, terminals, cores, coil forms, and wire) used in the manufacture.

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4.5.2 Test routine. - Sample units will be subjected to the preproduction inspection specified in table III, in the order shown. All sample units will be subjected to the inspection of group I. The sample units will then be divided into two groups of two units each. The sample units will then be subjected to the inspection for their particular group.

Table III. Preproduction inspection.

Examination or test	Requirement paragraph	Method paragraph
Group I (all sample units)		
Visual and mechanical examination (external).	3.1, 3.3 to 3.3.4, incl. 3.4 to 3.4.5.2, incl, 3.18, and 3.19	4.7.1.1
Electrical characteristics	3.5	4.7.2
Dielectric withstanding voltage	3.6	4.7.3
Insulation resistance (when applicable (see 3.1 and 6.1.2)).	3.7	4.7.4
Group II (2 sample units)		
Temperature rise (when applicable)	3.8	4.7.5
Terminal strength	3.9	4.7.6
Effects of soldering	3.10	4.7.7
High temperature	3.11	4.7.8
Polarity (focusing coils only)	3.12	4.7.9
Flux alinement (deflection coils only)	3.13	4.7.10
Electrical characteristics	3.5	4.7.2
Visual and mechanical examination (external)[1/].	3.1, 3.3 to 3.3.4, incl, 3.4 to 3.4.5.2, incl, 3.18, and 3.19	4.7.1.1
Group III (2 sample units)		
Vibration	3.14	4.7.11
Thermal shock	3.15	4.7.12
Immersion (grade 1 only)	3.16	4.7.13
Moisture resistance (grades 1 and 3 only)	3.17	4.7.14
Visual and mechanical examination (internal)[2/].	3.1, 3.4, 3.4.5.2, and 3.19	4.7.1.2

[1/] Marking will be considered defective only if it is illegible at the completion of any of the tests.
[2/] Only one sample unit need be disassembled for visual and mechanical examination (internal).

4.5.3 Failures. - No failures will be allowed.

4.6 Acceptance inspection. -

4.6.1 Inspection of product for delivery. - Inspection of product for delivery shall consist of groups A, B, and C.

4.6.1.1 Inspection lot. - An inspection lot, as far as practicable, shall consist of all the coils of the same grade and class, composition, and material, produced under essentially the same conditions, and offered for inspection at one time.

4.6.1.2 Resubmitted lots. - If an inspection lot is rejected, the supplier may replace it with a new lot, rework it to correct the defects, or screen out the defective units, and submit it again to inspection. Resubmitted lots shall be kept separate from new lots and shall be clearly identified as resubmitted lots. Resubmitted lots shall be inspected, using tightened inspection.

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4.6.1.3 Group A inspection. - Group A inspection shall consist of the examinations and test specified in table IV, and shall be made on the same set of sample units, in the order shown.

Table IV. Group A inspection.

Examination or test	Requirement paragraph	Method paragraph	AQL (percent defective)	
			Major	Minor
Visual and mechanical examination (external).	3.1, 3.3 to 3.3.4, incl, 3.4 to 3.4.5.2, incl, 3.18, and 3.19	4.7.1.1	1.0	4.0
Dielectric withstanding voltage (at atmospheric pressure)1/	3.6	4.7.3.1	1.0	---

1/ All defects are considered major.

4.6.1.3.1 Sampling plan. - Statistical sampling and inspection shall be in accordance with Standard MIL-STD-105 for ordinary inspection. The acceptable quality levels (AQL) shall be as specified in table IV. Major and minor defects shall be as defined in Standard MIL-STD-105 and in table V.

Table V. Classification of defects for visual and mechanical examination (external) in accordance with Standard MIL-STD-105.

Categories	Defects
Critical:	
1.....	None defined.
Major:	
101.....	Type, grade and class, not as specified.
102.....	Family not as specified.
103.....	Material nonconforming; evidence of unauthorized material used.
104.....	Parts, not as specified; not corrosion resistant, not thoroughly protected against corrosion.
105.....	Coils not mounted concentrically; not rigidly secured so as to prevent change in relative position of parts; damaged or excessively weak.
106.....	Threaded parts nonconforming; threads stripped, cracked, roughly finished.
107.....	Evidence of nonimpregnation, peeling, cracking, nonadherent areas.
108.....	External can shields, when required, not as specified; cannot be effectively mounted mechanically; cannot be grounded electrically by normal mounting means.
109.....	Solder terminals not effectively coated; cannot be readily soldered with tin-lead solder; not adequately identified; broken, cracked, loose, excessive amount of solder used.
110.....	Coil ends, end turns, and finish ends not secured in a manner so as to anchor them securely in place; lead lengths not as specified, and not supported, when necessary to prevent vibration.
111.....	Coils not thoroughly cleaned; evidence of dirt, grease, dust, foreign particles.
Minor:	
201.....	Marking nonconforming; illegible.

4.6.1.4 Group B inspection. - Group B inspection shall consist of the tests specified in table VI, in the order shown, and shall be made on sample units that have been subjected to and have passed group A inspection.

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Table VI. Group B inspection.

Test	Requirement paragraph	Method paragraph	AQL (percent defective)	Inspection level	
				Normal and tightened inspection	Reduced inspection
Dielectric withstanding voltage (at reduced barometric pressure).	3.6	4.7.3.2	4.0	L8	L6
Insulation resistance (when applicable (see 3.1 and 6.1.2)).	3.7	4.7.4			
Terminal strength	3.9	4.7.6			
Electrical characteristics	3.5	4.7.2	6.5	L7	L5

4.6.1.4.1 Sampling plan. - The sampling plan shall be in accordance with Standard MIL-STD-105 for small-sample inspection. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. For small-sample reduced inspection, procedure R-1 shall be used. The AQL and inspection levels shall be as specified in table VI.

4.6.1.4.2 Disposition of sample units. - Sample units which have passed all the group B inspection may be delivered on the contract or order, if the lot is accepted.

4.6.1.5 Group C inspection. - Group C inspection shall consist of the examinations and tests specified in table VII, in the order shown, and shall be made on sample units that have been subjected to and have passed group B inspection. Shipment shall not be held up pending results of the inspection.

Table VII. Group C inspection.

Examination or test	Requirement paragraph	Method paragraph
Polarity (focusing coils only)	3.12	4.7.9
Flux alinement (deflection coils only).	3.13	4.7.10
Thermal shock	3.15	4.7.12
Immersion (grade 1 only)	3.16	4.7.13
Visual and mechanical examination (internal)[1/].	3.1, 3.4, 3.4.5.2, and 3.19	4.7.1.2

[1/] Only one sample unit need be disassembled for visual and mechanical examination (internal).

4.6.1.5.1 Sampling plan. - Three sample units shall be selected from the first lot produced. Thereafter, three sample units shall be selected from each 500 units produced, or from each 3-months' production if less than 500 units were produced during that 3-month period. No failures will be allowed.

4.6.1.5.2 Disposition of sample units. - Sample units which have been subjected to group C inspection shall not be delivered on the contract or order.

4.6.1.5.3 Noncompliance. - If a sample fails to pass group C inspection, the supplier shall take corrective action on the materials or process, or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same conditions, with essentially the same materials, processes, etc, and which are considered subject to the same failure. Acceptance of the product shall be discontinued until corrective action, acceptable to the Government, has been taken. After the corrective action has been taken, group C inspection shall be repeated on additional sample units (all inspection, or the inspection which the original sample failed, at the option of the Government). Groups A and B

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inspection may be reinstituted; however, final acceptance shall be withheld until the group C reinspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure and the corrective action taken shall be furnished to the contracting officer.

4.6.2 Inspection of preparation for delivery. - Sample items and packs shall be selected and inspected in accordance with Specification MIL-P-116 to verify conformance with requirements in section 5 of this specification.

4.7 Methods of examination and test. -

4.7.1 Visual and mechanical examination. -

4.7.1.1 External. - Coils shall be examined to verify that the materials, external design and construction, physical dimensions, weight, type of mounting, marking, and workmanship are in accordance with the applicable requirements (see 3.1, 3.3 to 3.3.4, incl, 3.4 to 3.4.5.2 incl, 3.18, and 3.19).

4.7.1.2 Internal. - Coils shall be disassembled and examined to verify that the materials, internal design and construction, and workmanship are in accordance with the applicable requirements (see 3.1, 3.4, 3.4.5.2, and 3.19).

4.7.2 Electrical characteristics. - Coils shall be mounted by their normal mounting means, and the specified electrical characteristics (see 3.1 and 6.1.2) shall be determined (see 3.5, and 4.7.2.1 to 4.7.2.8, incl.).

4.7.2.1 Operating current and voltage. - Maximum operating current and voltage, as specified (see 3.1 and 6.1.2), shall be applied to each coil winding. The coil windings shall be observed for evidence of shorts, grounds, and sparking.

4.7.2.2 Direct-current resistance. - The direct-current (dc) resistance of coils shall be measured in accordance with method 303 of Standard MIL-STD-202.

4.7.2.3 Inductance. - Unless otherwise specified (see 3.1 and 6.1.2), the effective inductance of each coil winding shall be measured. Effective inductance shall be measured at the frequency and voltage level specified. (See 3.1 and 6.1.2.) For acceptance inspection, the measurement of effective inductance shall be referred to a comparison standard for correlation, when specified. (See 3.1, 4.7.2.2, and 6.1.2.) When true inductance is specified (see 3.1 and 6.1.2) the following test procedure, or equivalent, shall be used. A calibrated capacitor shall be used to tune the winding to resonance at several frequencies. These points will describe a straight line of added capacitance, one point of which shall be $\frac{1}{f_o}2$, as abscissa, versus the reciprocal of the frequency squared as ordinate, to be plotted, and the true inductance to be calculated by the following formula:

$$\text{True inductance} = KM$$

Where: $K = \frac{1}{4\pi^2}$

M = Slope of line representing added capacitance.

f_o = Self-resonant frequency of the coil at the abscissa of zero capacity.

4.7.2.4 Distributed capacitance. - The distributed capacitance of coils shall be determined, using the test circuit shown on figure 1, as follows:

(a) "N" as shown on figure 1 is not an electrical junction.
(b) All phases shall be tied to the core and grounded.
    1. Numbers 1, 2, and 3 shown on figure 1 are test-circuit terminals.
    2. A_1 and A_2 are the end connections of the vertical deflection coils, and B_1 and B_2 are the end connections of the horizontal deflection coils.
    3. Measure accurately the capacity between terminals 1 and 3 with switch "A" open and deflection coil disconnected. Use a "General Radio" Impedance Bridge No. 1650A, or equivalent, and record the value as C_2.

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4. Connect:

   (1) A_1 to terminal 1 of the test circuit.
   (2) N to terminal 2 of the test circuit.
   (3) A_2 to terminal 3 of the test circuit.

   (B_1 and B_2 shall not be connected.) Set the deflection coil in an insulated fixture approximately 6 inches above the bench.
5. Allow the coil to oscillate with switch "A" open, and adjust the audio oscillator to the same frequency as the coil. This condition will be indicated as a circle on the oscilloscope.
6. Close switch "A" and adjust the capacitor until the coil is oscillating at one-half of its original frequency. This condition will be indicated by a figure eight on the oscilloscope. The value of added capacity shall be recorded as C_1.
7. The distributed capacitance (C) shall be calculated by the following formula: $C = \frac{C_1}{3} - C_2$

   (where C_1 and C_2 are in micromicrofarads.)
8. Disconnect A_1 and A_2, and connect B_1 to terminal 1 of the test circuit and B_2 to terminal 3 of the test circuit.
9. Repeat steps 5, 6, and 7.

4.7.2.5 Deflection sensitivity (with accelerating voltage). - Deflection sensitivity (see 6.4.1) shall be measured in terms of the deflection factor (see 6.4.2) and the test shall be performed as specified (see 3.1 and 6.1.2).

4.7.2.6 Pincushion and barrel effect. - Pincushion and barrel effect (see 6.4.3) shall be measured for deviation exceeding the maximum allowable, as specified (see 3.1 and 6.1.2).

4.7.2.7 Perpendicularity. - Perpendicularity (see 6.4.4) shall be measured for deviation exceeding the maximum allowable, as specified (see 3.1 and 6.1.2).

4.7.2.8 Sweep straightness. - Sweep straightness (see 6.4.5) shall be measured for deviation exceeding the maximum allowable, as specified (see 3.1 and 6.1.2).

4.7.3 Dielectric withstanding voltage (see 3.6). - Coils shall be tested in accordance with 4.7.3.1, and when specified (see 3.1 and 6.1.2), in accordance with 4.7.3.2. Coil windings with special dielectric features (graded insulation) shall be subjected to the test potential specified for such windings. (See 3.1 and 6.1.2.)

4.7.3.1 At atmospheric pressure. - Coils shall be tested in accordance with method 301 of Standard MIL-STD-202. The following details and exceptions shall apply:

(a) Magnitude of test voltage - As specified. (See 3.1 and 6.1.2.) However, test voltages greater than 1,000 volts shall be applied gradually during preproduction inspection, at a rate not exceeding 500 volts per second.
(b) Nature of potential - Alternating current (ac).
(c) Duration of application of test voltage - For preproduction inspection, the test voltage shall be applied for not less than 60 seconds. For acceptance inspection, the test voltage shall be applied for a minimum of 5 seconds and not more than 60 seconds.
(d) Points of application of test voltage - Between each coil winding and the case or mounting hardware. Windings not being tested shall be grounded to the case or mounting hardware.
(e) Examinations after test - Coils shall be examined for evidence of arcing, flashover, breakdown of insulation, and other damage.

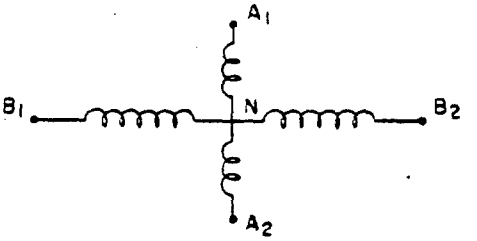
4.7.3.2 At reduced barometric pressure. - Coils designed for operation above 10,000 feet shall be tested as specified in 4.7.3.1 and in accordance with method 105 of Standard MIL-STD-202. The following details and exceptions shall apply:

(a) Method of mounting - By normal mounting means.
(b) Test-condition letter - As specified. (see 3.1 and 6.1.2.)

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(c) Magnitude of test voltage during subjection to reduced pressure - The root-mean-square (rms) test voltage shall be 1.2 times the maximum working voltage (see 3.1 and 6.1.2), but not less than 300 volts. Units with a working voltage less than 30 volts shall be tested at 100 volts.

(d) Examinations after test - Coils shall be examined for evidence of arcing, flashover, breakdown of insulation, and other damage.



TYPICAL SCHEMATIC OF A DEFLECTION COIL

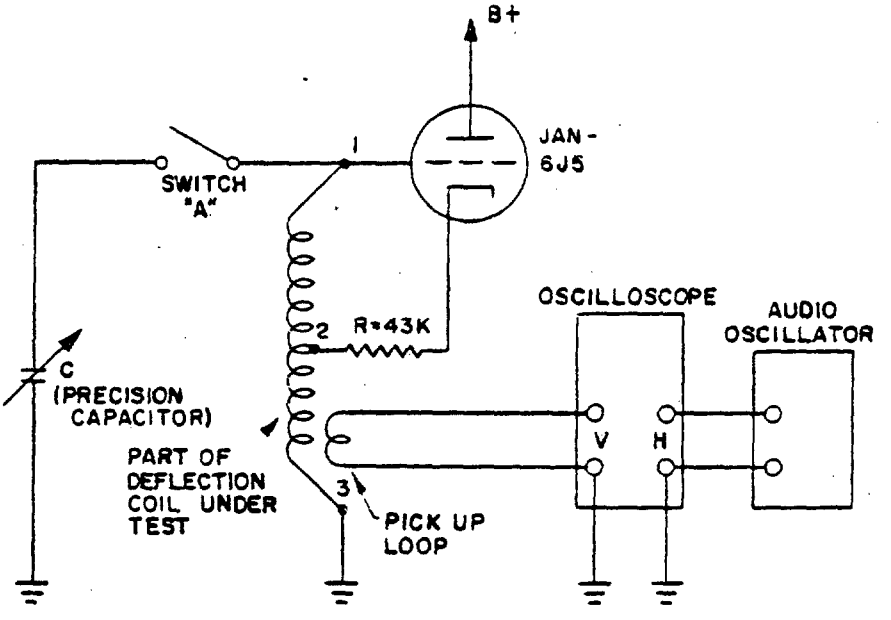


Figure 1. Test circuit for distributed capacitance.

4.7.4 Insulation resistance (when applicable (see 3.1 and 6.1.2)) (see 3.7). - Coils shall be tested in accordance with method 302 of Standard MIL-STD-202. The following details and exceptions shall apply:

(a) Test-condition letter - B, for preproduction inspection, except that for coils with a dielectric-withstanding test voltage less than 500 volts (see 4.7.3) test-condition letter A shall be used. For acceptance inspection, test-condition letter A, or a higher voltage shall be used; however, failure shall be based on test-condition letter A.

(b) Points of measurement - Measurements shall be made between mutually insulated parts. Each winding shall be tested with all other windings connected together, if applicable (see 3.1 and 6.1.2).

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4.7.5 Temperature rise (when applicable (see 3.1 and 6.1.2)). - The temperature rise of the windings shall be determined by the change-in-resistance method and shall be computed by the following formula:

$$\Delta T = \frac{R - r}{r} (t+234.5) - (T - t).$$

Where:

ΔT = Temperature rise in degrees centigrade above specified maximum ambient temperature. (See 3.1 and 6.1.2.)
R = Resistance of windings in ohms at temperature (T + ΔT).
r = Resistance of windings in ohms at temperature (t).
t = Specified initial ambient temperature in degrees centigrade.
T = Maximum ambient temperature in degrees centigrade at time of power shut-off. T shall not differ from t by more than 5° C.

The coils shall be conditioned at temperature (t) in a location free from drafts, with rated direct current (see 3.1 and 6.1.2) applied before resistance (r) is measured. When specified (see 3.1 and 6.1.2), an ac voltage of the frequency and magnitude specified (see 3.1 and 6.1.2) shall be superimposed on the direct current. When the resistance of the windings measured at 5-minute intervals remains constant, the temperature of the winding shall be considered as stabilized and resistance (R) shall be measured. If the method used for measuring resistance (R) requires the removal of power, the measurement shall be made within 30 seconds after the removal of power. (see 3.8).

4.7.6 Terminal strength. - Each terminal shall be subjected to a pull of 5 pounds unless otherwise specified (see 3.1 and 6.1.2), in a direction away from the unit and along the longitudinal axis of the terminal. The force shall be increased gradually to the specified magnitude and held at that value for at least 10 seconds (see 3.9).

4.7.7 Effects of soldering. - Terminals to be used for soldered connections shall be immersed, one at a time if possible, for 3 seconds in a pot of molten solder at a temperature of $350^{\circ}{}^{+10^{\circ}}_{-0^{\circ}}$ C. Each terminal shall be immersed to a point 1/4 inch from the nearest insulating material, or to one-half the exposed length of the terminal, whichever point is closer to the insulating material. After the test and as soon as the soldered terminals have returned to room ambient temperature, the coils shall be examined for softening of the insulation and loosening of the windings or terminals (see 3.10).

4.7.8 High temperature. - Coils shall be stored for 24 hours in a chamber maintained at a temperature of not less than the maximum operating temperature for the class. (See 1.2.1.2.2.) During the 24th hour and while still in the chamber at the above conditions, the coils shall be subjected to the dielectric-withstanding-voltage test (see 4.7.3.1) at a potential of 90 percent of that specified, and insulation resistance (when applicable (see 3.1 and 6.1.2)) shall be measured as specified in 4.7.4. After the test, coils shall be examined for softening of the insulation and loosening of the windings or terminals (see 3.11).

4.7.9 Polarity (focusing coils only). - A dc voltage shall be applied across the focusing coil as specified. (See 3.1 and 6.1.2.) The start terminal shall be connected to a negative voltage and a compass shall be inserted on the inside diameter of the coil, parallel to the axis of the coil. (See 3.12.)

4.7.10 Flux alinement (deflection coils only) (see 3.13). -

4.7.10.1 Vertical terminals. - Flux alinement for the vertical terminals of deflection coils shall be determined as follows: With a direct current as specified (see 3.1 and 6.1.2) obtained from a battery and current-limiting resistor, or an equivalent dc source, flowing into the positive vertical terminal and out of the negative vertical terminal, a compass (or search coil) placed at the geometric center of the coil shall indicate north in a direction as specified (see 3.1 and 6.1.2).

4.7.10.2 Horizontal terminals. - Flux alinement for the horizontal terminals of deflection coils shall be determined as follows: With a direct current as specified (see 3.1 and 6.1.2), or an equivalent dc source flowing into the positive horizontal terminal and out of the negative horizontal terminal, a compass (or search coil) placed at the geometric center of the coil shall indicate north in a direction which is at right angles (90°) to the direction obtained in 4.7.10.1.

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4.7.11 Vibration (see 3.14). - Coils shall be tested in accordance with method 201 of Standard MIL-STD-202. The following details and exceptions shall apply:

(a) Tests and measurements prior to vibration - Not applicable.
(b) Method of mounting - Coils shall be securely fastened by their normal mounting means or in a similar manner.
(c) Measurements during vibration - Not applicable.
(d) Measurements after vibration - The coils shall be subjected to the dielectric-withstanding-voltage test specified in 4.7.3.1, the insulation resistance (when applicable (see 3.1 and 6.1.2)) shall be measured as specified in 4.7.4, and the specified electrical characteristics (see 3.1 and 6.1.2) shall be determined (see 4.7.2).
(e) Examinations after vibration - Coils shall be examined for evidence of physical damage and winding discontinuity.

4.7.12 Thermal shock (see 3.15). - Coils shall be tested in accordance with method 107 of Standard MIL-STD-202. The following details and exceptions shall apply:

(a) Test-condition letter - A or B, as specified. (See 3.1 and 6.1.2.) For test-condition letter A, step 1 shall be performed at a temperature of -65° C; for steps 2 and 4, the period of time shall be 5 minutes, maximum.
(b) Measurements before cycling - Not applicable.
(c) Measurements after cycling - The coils shall be subjected to the dielectric-withstanding-voltage test (see 4.7.3.1) at a potential of 90 percent of that specified, applied for 5 seconds, and the insulation resistance (when applicable (see 3.1 and 6.1.2)) shall be measured as specified in 4.7.4.
(d) Examinations after test - Coils shall be examined for any impairment of the protective coating, evidence of leakage and physical damage, and corrosion affecting the mechanical or electrical operation.

4.7.13 Immersion (grade 1 only) (see 3.16). - Coils shall be tested in accordance with method 104 of Standard MIL-STD-202. The following details and exceptions shall apply:

(a) Test-condition letter - B.
(b) Measurements after final cycle - For acceptance inspection, coils shall be subjected to the dielectric-withstanding-voltage test (see 4.7.3.1) at a potential of 90 percent of that specified, and the insulation resistance (when applicable (see 3.1 and 6.1.2)) shall be measured as specified in 4.7.4.
(c) Examinations after test - Coils shall be examined for any impairment of the protective coating, evidence of leakage and physical damage, and corrosion affecting the mechanical or electrical operation.

4.7.14 Moisture resistance (grades 1 and 3 only) (see 3.17). - Coils shall be tested in accordance with method 106 of Standard MIL-STD-202. The following details and exceptions shall apply:

(a) Initial measurements - Not applicable.
(b) Polarization - Unless otherwise specified (see 3.1 and 6.1.2), polarization is applicable. The polarizing voltage shall be applied during steps 1 to 6 inclusive, between all windings normally maintained at a positive voltage and all other windings and the mounting hardware or case connected together. The polarizing voltage shall be positive with respect to the mounting hardware or case. For coils having only one winding or no mounting hardware, the polarizing voltage shall be applied as specified (see 3.1 and 6.1.2.).
(c) Loading voltage - Not applicable.
(d) Final measurements - After conditioning at high humidity, the coils shall be removed from the chamber and returned to the inspection conditions specified in 4.4. Within 30 minutes after such removal, the coils shall be subjected to the dielectric-withstanding-voltage test (see 4.7.3.1) at a potential of 90 percent of that specified, and the insulation resistance (when applicable (see 3.1 and 6.1.2)) shall be measured as specified in 4.7.4. Within 1 hour after the above tests are performed, the specified electrical characteristics (see 3.1 and 6.1.2) shall be determined (see 4.7.2). When specified (see 3.1 and 6.1.2), after a 4- to 24-hour drying period, coils shall again be subjected to the dielectric-withstanding voltage test

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(see 4.7.3.1) at a potential of 90 percent of that specified, the insulation resistance (when applicable (see 3.1 and 6.1.2)) shall be measured as specified in 4.7.4, and the specified electrical characteristics (see 3.1 and 6.1.2) shall be determined (see 4.7.2).

(e) Examinations after moisture resistance - Coils shall be examined for any impairment of the protective coating, evidence of physical damage, and corrosion affecting the mechanical or electrical operation.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging (see 6.1). -

5.1.1 Level A. - Coils shall be individually protected and unit-packaged in accordance with Specification MIL-P-116, as follows: Method III shall be used for grade 1. Unless otherwise specified (see 6.1), method IA, without the use of contact preservatives, shall be used for grades 2 and 3. When method IA-8 is used, the unit package shall be placed in a supplementary container conforming to Specification PPP-B-566 or PPP-B-676. Unless otherwise specified (see 6.1), five unit packages or a multiple thereof shall be further packaged in intermediate containers conforming to Specification PPP-B-566, class 1 of PPP-B-636, PPP-B-665, PPP-B-676. The gross weight of the intermediate container shall not exceed 10 pounds.

5.1.2 Level C. - Coils shall be afforded preservation and packaging in accordance with the supplier's normal commercial practice.

5.2 Packing (see 6.1). -

5.2.1 Level A. - Coils packaged as specified (see 6.1) shall be packed in overseas-type wirebound wood, wood-cleated fiberboard, wood-cleated plywood, nailed wood, fiber (class 2), or wood-cleated paper-overlaid boxes conforming to Specifications PPP-B-585, PPP-B-591, PPP-B-601, PPP-B-621, PPP-B-636, or PPP-B-676, respectively. Shipping containers shall have case liners conforming to Specification MIL-L-10547; the case liners shall be closed and sealed in accordance with the appendix thereto. Case liners for boxes conforming to Specification PPP-B-636 will not be required provided the center and edge seams and manufacturers' joints are sealed with tape, at least 1-1/2 inches wide, conforming to Specification PPP-T-76. Box closures and strapping shall be as specified in the applicable box specification or appendix thereto. Fiber boxes conforming to Specification PPP-B-636 may be banded with tape conforming to type IV of Specification PPP-T-97 and appendix thereto in lieu of steel straps. The gross weight of wood boxes shall not exceed 200 pounds; fiberboard boxes shall not exceed the weight limitations of the applicable box specification.

5.2.2 Level B. - Coils packaged as specified (see 6.1) shall be packed in domestic-type wirebound wood, wood-cleated fiberboard, wood-cleated plywood, nailed-wood, fiber (class 1 or 2, as specified (see 6.1)), or wood-cleated paper-overlaid boxes conforming to Specifications PPP-B-585, PPP-B-591, PPP-B-601, PPP-B-621, PPP-B-636, or PPP-B-676, respectively. Box closures shall be as specified in the applicable box specification or appendix thereto. The gross weight of wood boxes shall not exceed 200 pounds; fiberboard boxes shall not exceed the weight limitations of the applicable box specification.

5.2.3 Level C. - Coils packaged as specified (see 6.1) shall be packed in containers of the type, size, and kind commonly used for the purpose, in a manner that will insure acceptance by common carrier and safe delivery at destination. Shipping containers shall comply with the Uniform Freight Classification Rules, or regulations of other carriers as applicable to the mode of transportation.

5.2.4 General. - Insofar as possible and practical, exterior containers shall be uniform in shape and size, shall be of minimum cube and tare consistent with the protection required, and shall contain identical quantities of identical items.

5.3 Marking. - In addition to any special marking required by the contract or order, unit packages, supplementary containers, intermediate packages, and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129. (See 6.1.)

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6. NOTES.

6.1 Ordering data. -

6.1.1 For coil types covered by coordinated detail specifications. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Title, number, and date of the applicable detail specification and the complete type designation (see 1.2.1 and 3.1).
(c) Preproduction inspection (see 3.2):
  1. The laboratory at which preproduction inspection is to be performed (see 4.5).
  2. Sample, submission of data, and test routine, if other than that specified (see 3.2.1 and 4.5).
(d) Levels of preservation and packaging and packing, and applicable marking (see sect 5).
(e) Method of preservation and packaging of Specification MIL-P-116, if other than method IA (see 5.1.1).
(f) Number of unit packages if other than that specified in 5.1.1.
(g) Class of fiber (see 5.2.2).

6.1.2 For coil types not covered by coordinated detail specifications. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type designation, covering the component, grade and class, and family, minus the identification number (see 1.2.1 to 1.2.1.3, incl).
(c) Preproduction inspection (see 3.2):
  1. The laboratory at which preproduction inspection is to be performed (see 4.5.)
  2. Sample, submission of data, and test routine, if other than that specified (see 3.2.1 and 4.5).
(d) Applicable requirements for use of fungus-inert materials (see 3.3.)
(e) Type of magnet wire required (see 3.3.5 and table II).
(f) Type of varnish impregnant required (see 3.3.6 and table II).
(g) Applicable drawings covering mounting and other physical dimensions (see 3.4).
(h) Required weight of the coil (see 3.4.1).
(i) Type of mounting required (see 3.4.2).
(j) Whether external can shields are required (see 3.4.4).
(k) Length of leads from winding ends (see 3.4.5.2).
(l) Whether insulation resistance is applicable initially (see 3.7 and 4.7.4), and if so:
  1. Insulation-resistance values for grades 2 and 3 (see 3.7).
(m) Whether temperature rise is applicable (see 3.8 and 4.7.5), and if so:
  1. The maximum allowable temperature rise.
(n) Whether insulation resistance is applicable after the following:
  1. High temperature (see 3.11 and 4.7.8).
  2. Vibration (see 3.14 and 4.7.11).
  3. Thermal shock (see 3.15 and 4.7.12).
  4. Immersion (see 3.16 and 4.7.13).
  5. Moisture resistance (see 3.17 and 4.7.14).
(o) Polarity required for focusing coils (see 3.12).
(p) Flux alinement for deflection coils (see 3.13).
(q) Other marking information (see 3.18).
(r) Required electrical characteristics, including values and tolerances (see 3.5 and 4.7.2):
  1. Test frequencies at which electrical characteristics are to be measured.
  2. Maximum operating current and voltage to be applied to each coil winding (see 4.7.2.1.)
(s) For inductance test (see 4.7.2.3):
  1. Whether the effective inductance of each coil winding shall be measured.
  2. For acceptance inspection, whether measurement of effective inductance is to be referred to a comparison standard.
  3. Whether true inductance is required.

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(t) Test procedures required for the following electrical characteristics:
1. Deflection sensitivity (with accelerating voltage) (see 4.7.2.5).
2. Pincushion and barrel effect (see 4.7.2.6).
3. Perpendicularity (see 4.7.2.7).
4. Sweep straightness (see 4.7.2.8).

(u) Test potential for coil windings with special dielectric features (graded insulation) (see 4.7.3).
(v) Magnitude of test voltage for dielectric-withstanding-voltage (at atmospheric pressure) test (see 4.7.3.1).
(w) Whether reduced-barometric-pressure test is required (see 4.7.3.2), and if so:
1. Test-condition letter.
2. Maximum working voltage.

(x) Whether coil windings are connected together for insulation-resistance measurements (see 4.7.4).
(y) For temperature-rise test (see 4.7.5):
1. Maximum ambient temperature.
2. Rated direct current.
3. Whether ac voltage is required, and if so, the required frequency and magnitude.

(z) Magnitude of terminal pull, if other than 5 pounds. (see 4.7.6.)
(aa) For polarity test, the method of applying dc voltage for focusing coils (see 4.7.9).
(bb) Direct current and direction of north for flux alinement. (see 4.7.10.1 and 4.7.10.2).
(cc) For thermal-shock test, whether test-condition leter A or B is applicable (see 4.7.12).
(dd) For moisture-resistance test (see 4.7.14):
1. Whether polarization is applicable, and if so, the method of applying voltage for coils having only one winding or no mounting hardware.
2. Whether additional final measurements are to be made after a 4- to 24-hour drying period.

(ee) Levels of preservation and packaging and applicable marking (see sect 5).
(ff) Method of preservation and packaging of Specification MIL-P-116, if other than IA (see 5.1.1).
(gg) Number of unit packages if other than that specified in 5.1.1.
(hh) Class of fiber (see 5.2.2).

6.2 Preproduction inspection. - Information pertaining to preproduction inspection of products covered by this specification should be obtained from the procuring activity for the specific contracts involved (see 3.2).

6.3 Assignment of type designation. - Complete type designations, including the identification numbers (see 1.2.1.4), will be assigned by the Armed Services Electro-Standards Agency (ASESA), Fort Monmouth, N. J., upon request of the Army, the Navy, or the Air Force, only when a coordinated detail specification has been established.

6.4 Definitions of dynamic electrical parameters. - Dynamic electrical parameters are defined in 6.4.1 to 6.4.5, inclusive.

6.4.1 Deflection sensitivity. - Deflection sensitivity is defined as the number of radians (or decimal fraction thereof) by which the electron beam is deflected per milliampere of current caused to flow through the coil windings when the cathode-ray tube anode voltage is as specified in the contract or order. (One radian is equivalent to 54-degree deflection angle.) When specified in this manner, the deflection sensitivity of the coil is correct for any cathode-ray tube type used (see 4.7.2.5).

6.4.2 Deflection factor. - Deflection factor is defined as the current (in milliamperes) required to produce a unit deflection; a unit deflection is one radian of the screen of the cathode-ray tube. Deflection factor is the reciprocal of the deflection sensitivity, and is expressed as milliamperes per radian (where one radian is equivalent to 54-degree deflection angle) (see 4.7.2.5).

6.4.3 Pincushion and barrel effect. - Pincushion and barrel effect is defined as the maximum allowable deviation from a straight line on the sides of a 6-inch (or other specified size (see figure 2)) square raster, when the deflection coil is used on the specified cathode-ray tube operating at the conditions specified in the contract or order (see 4.7.2.6).

6.4.4 Perpendicularity. - Perpendicularity is defined as the maximum allowable deviation from a 90° angle on a square raster, as shown on figure 3 (see 4.7.2.7).

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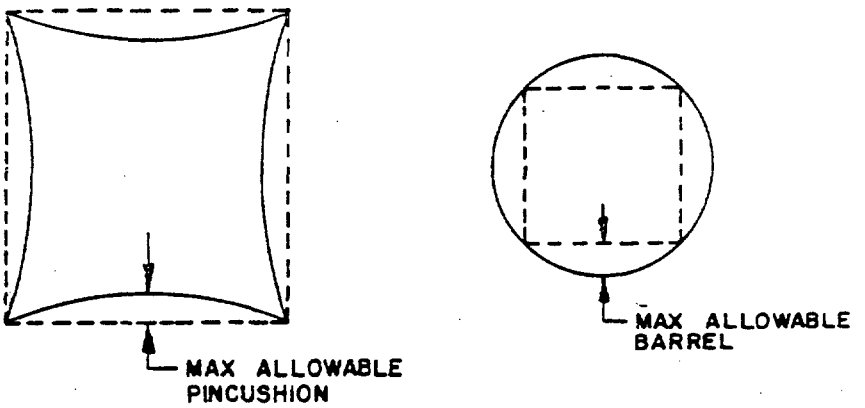


Figure 2. Configuration of pincushion and barrel effect.

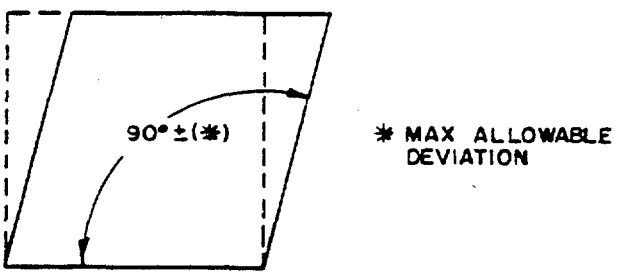


Figure 3. Example of perpendicularity.

6.4.5 Sweep straightness (see 4.7.2.3). -

6.4.5.1 Single-axis coils. - Sweep straightness on single-axis coils is defined as the maximum allowable deviation from a straight line when the coil is operating with specified sweep voltages across its terminals and operating with the cathode-ray tube specified in the contract or order.

6.4.5.2 Double-axis coils. - Sweep straightness on double-axis coils is defined as the maximum allowable deviation from a straight line when one of the two coils is operating with the specified sweep voltages across its terminals with the cathode-ray tube specified in the contract or order, and the inactive coil (other axis) having a specified termination (normal loading); or it can be a measured induced crosstalk voltage across the terminated inactive coil, with the same voltage and frequency applied to the active coil.

6.5 Design information. - The following design information should be included in the procurement document to insure that coils which best fit the application are obtained:

(a) Description of the type of scan:
1. Minimum and maximum sweep rates.
2. Duty ratios.
3. Operating frequencies.
4. Waveforms.

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(b) Limitations of proposed driving circuit:
1. Power supply voltages.
2. Peak currents.
3. Type of drive to be employed.

(c) Cathode-ray tube requirements:
1. Deflection angle.
2. Accelerating potential.
3. Accuracy of presentation.
4. Resolution.

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

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